

SUMMARY

Senior Software Engineer with 15+ years of experience in JavaScript, TypeScript, Python, Go, and Rust, building scalable backend systems with Fastify, Express, Django, and FastAPI on AWS and GCP. Architected microservices, event-driven systems, and cloud deployment frameworks at companies like Coursedog, Telus, and HumanAPI — processing 1M+ daily interactions, stabilizing 15+ university integrations, and cutting deployment times by 80%. Integrating LLM capabilities into client-facing tools to reduce support overhead and improve self-service onboarding.

SKILLS

Languages: JavaScript, TypeScript, Python, Go, Rust, SQL.

Infrastructure: AWS, GCP, Kubernetes, Docker, Terraform, Pulumi, GitHub Actions, CloudFormation, HashiCorp Vault, Nginx, Serverless.

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch, Supabase, Firebase, Amazon RDS, SQLite.

Message Queues & Streaming: Kafka, RabbitMQ, Redis Streams, Socket.io, WebSocket, ZeroMQ.

Testing: Jest, Vitest, Playwright, Tap, Nock, React Testing Library.

Frameworks: Fastify, Express, Django, FastAPI, Hono, React, Next.js, Svelte, Vue.js, Flask.

Architecture & Patterns: Microservices, Event-Driven Architecture, CQRS, Event Sourcing, Saga Pattern, API Gateway, Module Federation, Microfrontends, Domain-Driven Design.

Observability: Prometheus, Grafana, Jaeger, OpenTelemetry, Distributed Tracing.

WORK EXPERIENCE

Senior Software Engineer. Coursedog. New York, NY

Mar 2023 - Present

- Reduced ERP integration support tickets from ~6/week to 1-2/week by implementing configurable retries and dynamic header injection in TypeScript, which stabilized sync reliability across 15+ university integrations.
- Reduced data engineer questions by ~40% and accelerated client onboarding across 300+ institutions by leading the end-to-end Data Lineage feature (RFC, API, and UI) with an LLM-powered chat interface using custom `@sisField` and `@sisEntity` TypeScript decorators and Vue.js, which replaced a stale Salesforce document with live, self-updating source field traceability.

Senior Engineer. Diligent (Previously Galvanize). Vancouver, BC

Oct 2021 - Mar 2022

- Cut test execution from ~45 to ~18 minutes by parallelizing Jest across available resources in Node.js, which unblocked the release cycle for a product serving 100K+ daily users.
- Reduced flaky test failures by establishing testing standards and distributing the Jest suite across CI resources, which accelerated PR review cycles using Node.js.

Staff Software Engineer. Sharecare (Previously Doc.ai). San Francisco, CA

Mar 2022 - Mar 2023

- Cut API calls by 50% and raised test coverage from 80% to 95% by implementing React Context, react-query, and a shared Storybook component library across 5 projects, which eliminated frontend duplication and accelerated feature delivery on a mobile health platform.
- Cut feature delivery time from ~3 days to under 1 day by eliminating duplicated CSS and components across 5 projects using a shared Storybook design system with React Context and react-query.

Senior Technology Architect. Telus. Vancouver, BC

Mar 2019 - Jul 2020

- Cut deployment time from ~4 hours to ~30 minutes by building a Fastify-based application framework with bundled secrets management, linting, and config as npm packages, which replaced error-prone template copy-pasting with npm create/update workflows and achieved 95% automated secret rotation across 15+ teams.
- Centralized authentication for 20+ services by building an auth-aware API gateway with TypeScript, Fastify, and OAuth2, which enforced access control and standardized auth forwarding.
- Enabled full audit trails and crash recovery by designing an event-sourced CI/CD pipeline with CQRS in TypeScript and Go on GCP/AWS, which processed ~30+ deployments/day with time-travel debugging.

Senior Engineer. HumanAPI. San Mateo, CA

Jul 2015 - Jun 2018

- Cut search time from ~500ms to ~150ms by replacing slow MongoDB queries with Elasticsearch, which eliminated the support ticket backlog for missing clinics across 15+ provider integrations.
- Reduced sync failures from ~40/week to near-zero by normalizing health data across 15+ providers using a Node.js microservices layer with RabbitMQ, which eliminated customer-reported data issues.
- Raised test coverage from 65% to 92% by replacing brittle mocks with integration tests in Node.js and Jest, which cut release verification from 2+ hours to automated confidence.

Lead Architect and Software Engineering Consultant. MobileLive. Vancouver, BC

Apr 2021 - Oct 2021

- Gave 4+ teams independent deployment capability by migrating a monolithic frontend to microfrontends with bolsterJS, a Webpack Module Federation abstraction that reduced setup complexity by 75% and enabled independent frontend ownership.

Senior Lead Engineer. CTO.ai. Vancouver, BC

Aug 2020 - Apr 2021

- Gave leadership real-time project health visibility by building a DORA metrics dashboard with a custom GitHub App in TypeScript and React, which reduced employee churn and became the company's core product.
- Eliminated manual DevOps workflows by building a Go event queue processing 50+ daily Slack-driven commands with async responses, which freed engineers from context-switching for routine operations.

Principal Lead Engineer. Selfie Inc.. New York, NY

May 2013 - Jul 2015

- Brought timeline loads from ~1s to ~200ms and scaled capacity 3x by implementing CQRS with async queues and fan-out caching, which stopped user churn from celebrity-driven traffic spikes.
- Reduced deployment time by 75% by automating the pipeline across 5+ microservices with Capistrano and an in-house CI server, which eliminated manual deploys and enabled same-day hotfixes.

EDUCATION

Pontifical Bolivarian University, Colombia, Antioquia, Medellin. B.Sc. Electronics Engineer

Julio 2002 - Abril 2007

University of the Andes, Colombia, Bogota D.C.. M.Sc. in Computer Engineering

Julio 2011 - Febrero 2014

OTHER

Languages: English (Native), Spanish (Native).